Investigating Falling Droplet Behavior
530.777 Multiphase FLows Project 1

Weixuan Li

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This study investigates the dynamics of a two-dimensional falling droplet under the influence of body forces using numerical simulations. Utilizing the *CodeC3-frt-st-RK3.m* developed by Gretar Tryggvason, we analyze the evolution of the droplet's aspect ratio and centroid velocity over time. The effects of key nondimensional numbers, specifically the Eötvös (Eo) and Ohnesorge (Oh) numbers, on droplet deformation and breakup behavior are systematically examined. Grid independence tests validate the accuracy of the simulations, ensuring reliable results. The findings reveal distinct deformation modes influenced by varying Eo and Oh values, with comparisons drawn against published data from Han et al. [1] to benchmark the computational approach. This comprehensive analysis enhances the understanding of the interplay between body force, surface tension, and viscous forces governing the behavior of falling droplets, providing valuable insights for both natural phenomena and engineering applications.

Contents

1 Introduction

In multiphase flow research, the study of a falling drop is of significant interest due to its relevance in a wide range of natural and industrial processes. Understanding the behavior of falling drops is crucial for applications ranging from weather forecasting to advanced material science. For example, in meteorology, raindrop formation and dynamics influence weather patterns and precipitation rates, impacting everything from agriculture to urban water management [2]. The way droplets fall, coalesce, or break apart in the atmosphere can significantly affect rainfall intensity and distribution, making it an important factor in climate models and weather prediction systems [3].

Industrially, falling droplets are essential in spray cooling systems, where they are used to efficiently dissipate heat from surfaces in power plants, electronics, and manufacturing processes [4]. The ability to control droplet size and fall behavior directly influences the cooling efficiency and safety of such systems. In another domain, pharmaceutical industries encounter falling droplets in drug delivery systems, where droplets must be precisely controlled to deliver medication in aerosol or liquid form [5]. The mechanics of droplet formation and behavior play a key role in determining how effectively the medication reaches its target in the body.

These examples highlight the importance of studying falling droplets in various fields. The complex interactions between the droplet, the surrounding fluid, and gravity provide a rich framework for exploring phenomena such as droplet deformation, fragmentation, and interactions with turbulent flows. By investigating these processes, we can gain a deeper understanding of multiphase flow dynamics, which is critical for improving efficiency and innovation across multiple industries.

The oldest known scientific exploration of falling droplets is often credited to Isaac Newton, who addressed various aspects of fluid mechanics in his *Philosophiae Naturalis Principia Mathematica* [6]. In this work, Newton also touched upon the refraction of light in raindrops, a phenomenon later linked to rainbow formation. More focused investigations into the dynamics of falling droplets emerged much later, notably in the 19th and early 20th centuries. A.M. Worthington was among the pioneers in this field, publishing an influential paper in the *Philosophical Transactions of the Royal Society of London*, where he described the observation of oscillations in falling drops [7]. In a subsequent paper, Worthington supplemented this work by delving into the early stages of droplet oscillation, documenting the diverse forms and behaviors of liquids, such as water and mercury, as they fall vertically onto a horizontal surface [8].

A recent paper can be referred to Ref. [9], which explores the dynamics of water droplets impacting non-wetting surfaces, such as those found in non-stick frying pans. It describes how droplets initially spread upon impact, then retract due to surface tension, forming a jet that shoots upwards, which can create a second rebound force. Remarkably, under certain conditions, this second force can be even stronger than the initial impact.

This study investigates the dynamics of a two-dimensional falling drop subjected to body forces. Through numerical simulations conducted using *CodeC3-frt-st-RK3.m* developed by Gretar Tryggvason [10], we analyze the evolution of the drop's aspect ratio and centroid velocity over time. The influence of key nondimensional numbers, specifically the Eötvös (Eo) and Ohnesorge (Oh) numbers, on the drop's deformation and breakup behavior is systematically examined. Grid independence tests are performed to ensure the accuracy and reliability of the simulations. Additionally, the results are compared against published data from Ref. [1] to validate the computational approach. This comprehensive investigation provides valuable insights into the complex interactions between body force, surface tension, and viscous forces governing the behavior of falling droplets, with implications for both natural phenomena and engineering applications.

2 Methodology

2.1 Numerical Method

The governing equations, incompressible Navier–Stokes equations, of the fluid dynamics are presented as follows

$$\nabla \cdot \mathbf{u} = 0, \tag{1}$$

$$\rho \left[\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) \right] = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g} \tag{2}$$

where $\mathbf{u}$ is the fluid velocity vector, p is the pressure, and ρ and μ are density and dynamic viscosity respectively; $\mathbf{g}$ is the body force acceleration and represented by a_z in this study. The simulations were conducted using *CodeC3-frt-st-RK3.m* developed by Gretar Tryggvason [10]. This code employs a finite volume method to solve the Navier-Stokes equations and utilizes a front-tracking technique to model multiphase flow dynamics.

2.2 Non-dimensional Analysis

Considering a drop falling through the ambient fluid, various parameters influence the behavior of the falling drop. These parameters include the density of the ambient fluid (ρ_o) and liquid (ρ_d), the dynamic viscosity of the ambient fluid (μ_o) and liquid (μ_d), the surface tension coefficient (σ), the drop diameter (D), evolution time (t), and the body force acceleration (a_z). The general form of the equation governing this process can be written as:

$$t = f(\rho_o, \rho_d, \mu_o, \mu_d, \sigma, D, a_z)$$

The key non-dimensional numbers are:

Eötvös number (Eo)

$$Eo = \frac{a_z \Delta\rho D^2}{\sigma}$$

Ohnesorge number (Oh)

$$Oh = \frac{\mu_d}{\sqrt{\rho_d D \sigma}}$$

Density ratio (ρ^*)

$$\rho^* = \frac{\rho_d}{\rho_o}$$

Viscosity ratio (μ^*)

$$\mu^* = \frac{\mu_d}{\mu_o}$$

Dimensionless time (t^*)

$$t^* = \frac{t}{\sqrt{D/a_z}}$$

Eötvös number characterizes the ratio of body forces to surface tension forces in a fluid. Ohnesorge number is the ratio of viscous forces to surface tension and inertial forces in a fluid. Considering the natural phenomenon of a raindrop falling through air (diameter $D = 2mm$, density of air $\rho_o = 1.2kg/m^3$, density of drop $\rho_d = 1000kg/m^3$, viscosity of air $\mu_o = 1.81 \times 10^{-5} Pa \cdot s$, viscosity of drop $\mu_d = 1.002 \times 10^{-3} Pa \cdot s$, surface tension coefficient $\sigma = 72.8 \times 10^{-3} N/m$), the body force acceleration should be gravity acceleration $a_z = 9.8m/s^2$, and then the values of the dimensionless numbers are:

- **Eötvös number**: $Eo = \frac{a_z \Delta\rho D^2}{\sigma} = \frac{9.8\,\mathrm{m/s^2} \times 998.8\,\mathrm{kg/m^3} \times (0.002\,\mathrm{m})^2}{72.8 \times 10^{-3}\,\mathrm{N/m}} = 0.538.$
- **Ohnesorge number**: $Oh = \frac{\mu_d}{\sqrt{\rho_d D \sigma}} = \frac{1.002 \times 10^{-3}\,\mathrm{Pa \cdot s}}{\sqrt{1000\,\mathrm{kg/m^3} \times 0.002\,\mathrm{m} \times 72.8 \times 10^{-3}\,\mathrm{N/m}}} = 0.00263.$
- **Density ratio**: $\rho^* = \frac{\rho_d}{\rho_o} = \frac{1000\,\mathrm{kg/m^3}}{1.2\,\mathrm{kg/m^3}} = 833.$
- **Viscosity ratio**: $\mu^* = \frac{\mu_d}{\mu_o} = \frac{1.002 \times 10^{-3}\,\mathrm{Pa \cdot s}}{1.81 \times 10^{-5}\,\mathrm{Pa \cdot s}} = 55.4.$

In the analysis of limiting cases, when the Eötvös number (Eo) and Ohnesorge number (Oh) are low, surface tension and viscous forces dominate. This typically corresponds to small droplets where body effects are minimal (low Eo), and viscous forces are significant relative to inertial forces (low Oh). Under these conditions, droplets tend to remain stable and nearly spherical, as surface tension minimizes the surface area, and deformation is limited. In contrast, when the Eötvös number is high, body forces become more influential, leading to droplet elongation, deformation, or potential fragmentation. If the Ohnesorge number is high, viscous forces resist deformation and prevent droplet breakup even under the influence of significant body forces. However, with low Ohnesorge numbers and moderate to high Eötvös numbers, the droplet may be more susceptible to deformation and breakup due to the reduced resistance to external forces. These dimensionless numbers provide valuable insight into predicting droplet stability, deformation, and fragmentation under various physical conditions.

2.3 Numerical Setup

We consider a 2D drop falling from the stationary state by the effect of body force. The computational domain is 5×15 times the diameter of the drop. The four boundaries are non-slip boundaries. The schematic illustration of computational setup is shown in Fig. 1.

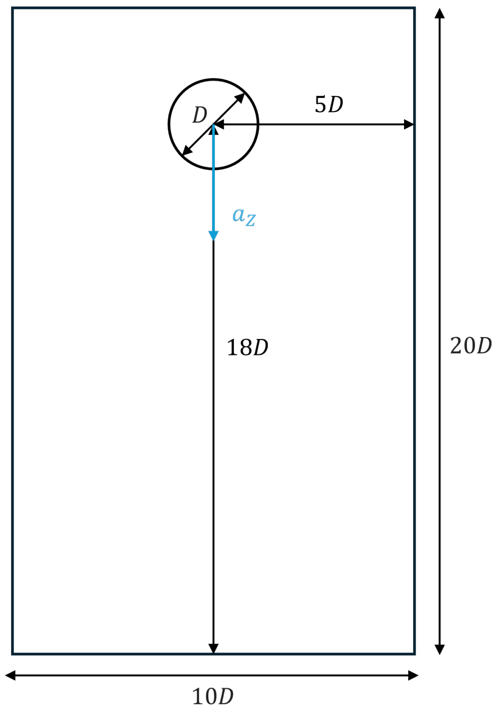


Figure 1: Schematic illustration of computational setup.

3 Results and Discussions

3.1 Grid Independence Test

First, grid independence tests were performed to validate the numerical results. We assume that the diameter of the drop is $D = 2$, the surface tension coefficient is $\sigma = 300$, the density of the ambient fluid is $\rho_o = 1$, and the density ratio is $\rho_d/\rho_o = 10$. The Eötvös number (Eo) and the Ohnesorge number (Oh) are controlled by the body force a_z and the viscosity μ, respectively. For the grid independence tests, we chose $Eo = 12$ and $Oh_d = Oh_o = 0.05$. From these parameters, the body force is calculated as

$$a_z = \frac{Eo}{0.12} = 100$$

The viscosities of the drop and the ambient fluid are

$$\mu_d = Oh_d\sqrt{6000} = 3.87$$

$$\mu_o = Oh_o\sqrt{600} = 1.22$$

Three different grid resolutions were used: 128×192, 192×288, and 256×384. The contours of the drop falling and breaking up computed using grid resolutions of 192×288 and 256×384 are shown in Fig. 2. The contours are plotted at every $\Delta t^* = 0.3369$.

The aspect ratios and the centroid velocities of the results with different resolutions are plotted in Fig. 3.

The contours, aspect ratios, and centroid velocities are similar for grid resolutions of 192×288 and 256×384. By comparing the results across the three resolutions, it can

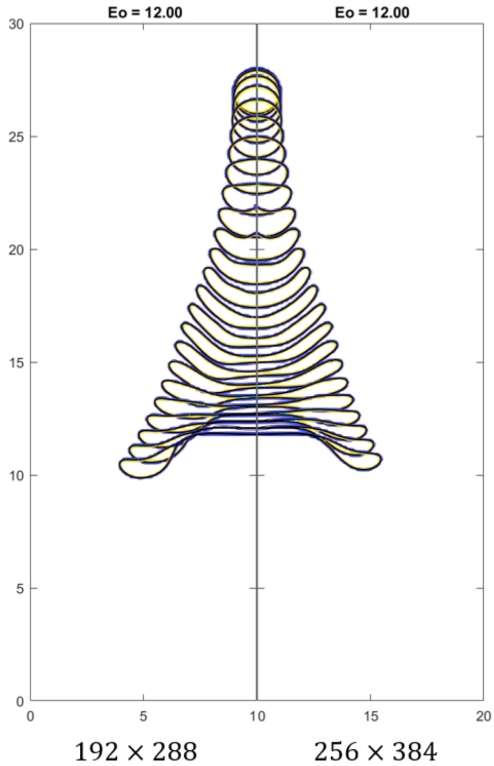


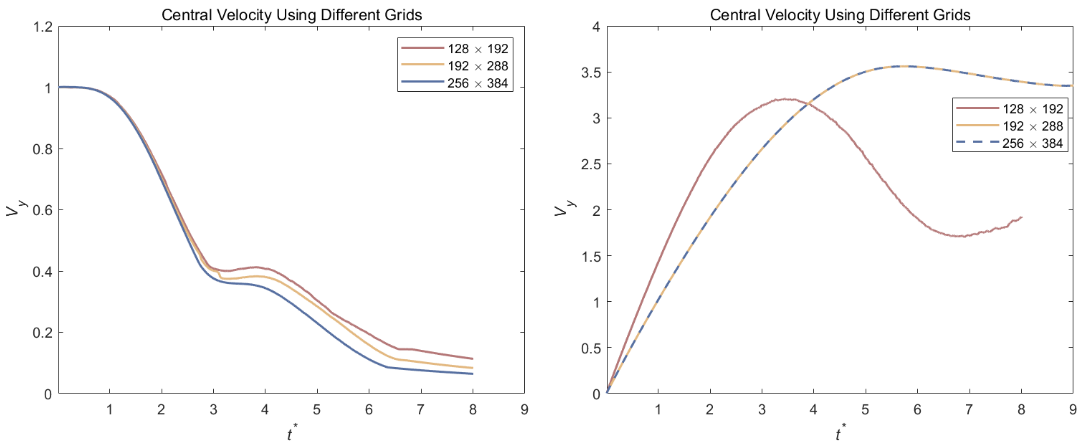
Figure 2: Grid independence test. The falling and breaking up of a drop computed using a 192×288 grid (left) and a 256×384 grid (right).

be observed that the outcomes converge to those obtained with the 256×384 grid as the resolution increases. Therefore, we conclude that when the grid resolution is larger than or equal to 192×288, the results are grid-independent and can be considered accurate.

3.2 Effects of Eötvös Number at Small Ohnesorge Number

When the Ohnesorge number (Oh) is low and surface tension significantly surpasses viscous stresses, Ohnesorge number plays a minimal role in drop breakup, making the Eötvös number (Eo) the primary governing factor [1]. In this section, we present findings for various Eo values under conditions of low Ohnesorge number, i.e. $Oh = 0.05$. The different breakup behaviors observed are determined by the Eo, which measures the balance between pressure forces and surface tension. The surface tension coefficient, density and viscosity of the drop and the ambient fluid are the same as those in grid independence test.

The drop contours are shown in Fig. 4 when $Eo = 6, 12, 28.8$. In scenario (a), with Eo=6, the drop begins to fall, flattening at the rear while the front remains rounded. After this initial shape change, the drop stabilizes, maintaining a constant form. Increasing Eo to 12 in scenario (b) results in more significant deformation. Initially resembling the shape in a, the drop's rear becomes progressively more convex, gradually transforming into a thin, disk-like form. Subsequently, the thickness near the symmetry axis continues to thin, causing most of the fluid to migrate toward the drop's perimeter. This leads to the formation of a backward-facing bag as the center of the front



(a) Aspect ratio (b) Centroid velocity

Figure 3: Grid independence test. The aspect ratios and the centroid velocities versus non-dimensional time computed using different resolutions: 128×192, 192×288, and 256×384 are shown.

surface elevates. At this stage, the fluid primarily resides in an annular rim.

Finally, for Eo=28.8 in scenario (c), another deformation mode is observed. Initially, the drop's evolution mirrors that of previously discussed cases. However, as time progresses, notable differences emerge. The indentation at the top of the drop becomes increasingly pronounced, preventing the drop from thinning into a disk-like structure. Instead, the perimeter of the drop bends downstream, extending into a slender film. Meanwhile, the central region of the drop retains a convex form, and the thickness along the symmetry axis ceases to diminish.

Fig. 5 illustrates aspect ratios α and the centroid velocities V_y plotted against the dimensionless time t^* for the drops depicted in Fig. 4, respectively. The aspect ratio of the drop, defined as the thickness along the symmetric axis relative to the width between the left and right edge points, exhibits distinct behavior depending on the Eötvös number (Eo). For $Eo = 6$, the aspect ratio gradually decreases over time, reflecting the minimal deformation of the drop. As Eo increases to 12, the aspect ratio diminishes at a more accelerated rate, and this trend becomes even more pronounced at $Eo = 28.8$. Specifically, for $Eo = 12$, the onset of bag breakup causes a rapid reduction in the aspect ratio. Similarly, at $Eo = 28.8$, the bending of the drop's perimeter downstream leads to a swift decrease in the aspect ratio. After these rapid deformations, the aspect ratio begins to stabilize as the drop's shape becomes defined.

The initial slopes of the velocity curves are identical across different Eötvös numbers since the velocity is non-dimensionalized by $\sqrt{D/a_z}$. Following the initial acceleration phase, the deformation of the drop, influenced by Eo, governs the subsequent velocity behavior. In Fig. 4, drops with lower Eo values undergo less deformation and, consequently, move faster than those with higher Eo. Specifically, the drop with the smallest Eo ($Eo = 6$) asymptotically reaches a steady velocity (blue line). In contrast, other drops begin to decelerate as they start to deform. For drops experiencing bag breakup like the $Eo = 12$ drop, once the bag forms, the velocities decrease sharply. The drop with $Eo = 28.8$ experiences rapid deceleration when stretched perpendicular to the flow and slightly speeds up as the thin films drawn from its edges fold back

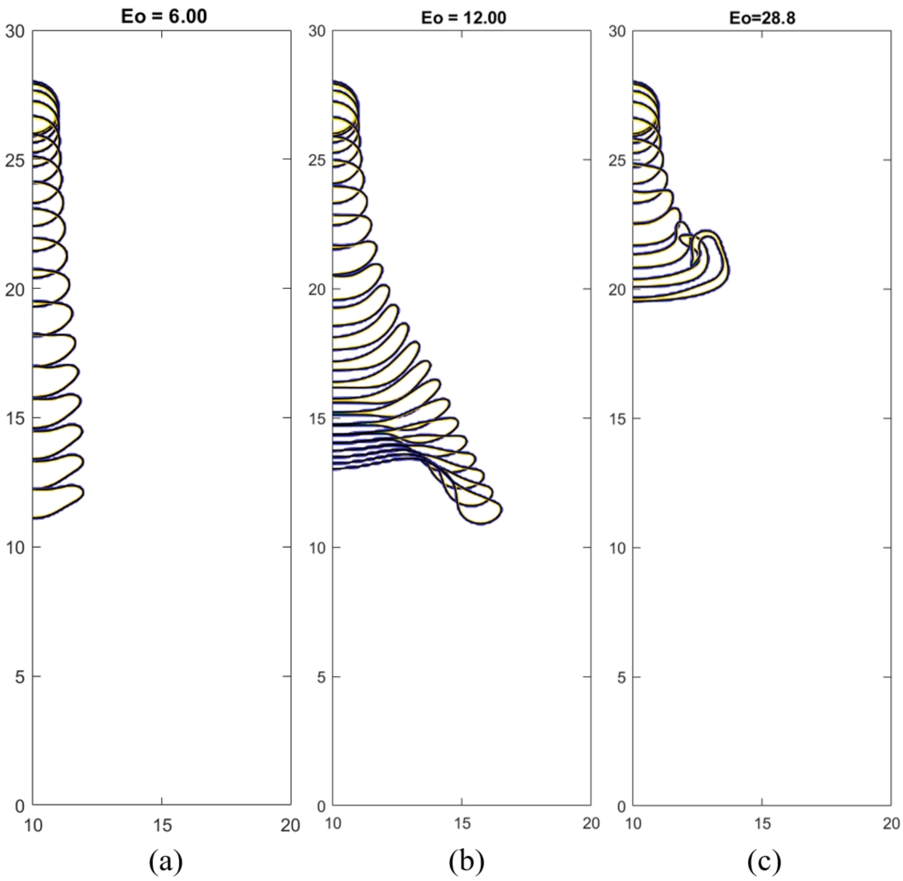


Figure 4: Effects of Eo on the falling and breaking up of a drop. The time interval is $\Delta t^* = 0.3369$. The last interface is plotted at $t^* = 6$ (a), $t^* = 8$(b) and $t^* = 4$ (c).

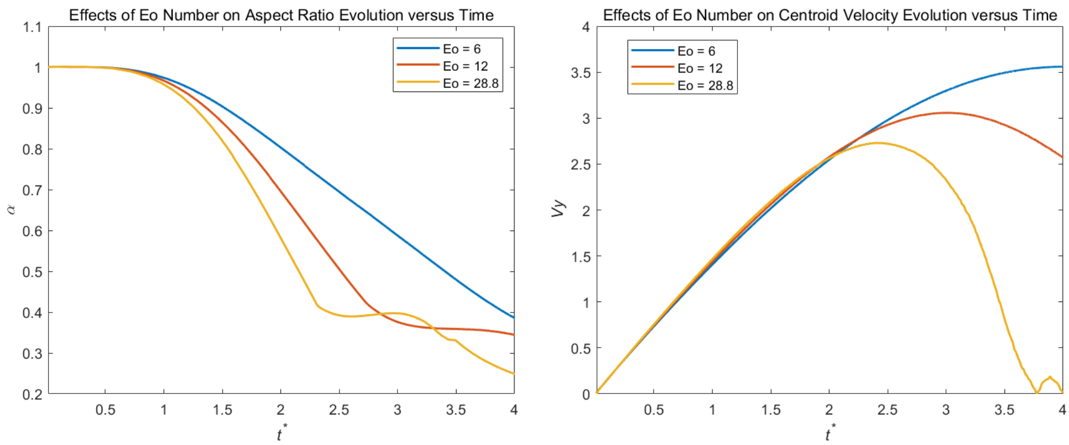
toward the central axis.

3.3 Effects of Ohnesorge Number

With the same surface tension coefficient, density and viscosity of the drop and the ambient fluid, and fixed Eötvös number $Eo = 28.8$, the effects of Ohnesorge number are investigated in this section. We compare the results between $Oh_o = Oh_d = 0.05, 0.07, 0.1$. The contours are shown in Fig. 6.

In Fig. 6, the Eötvös number (Eo) is set to be 28.8, and the evolution of the drops is presented for the three Ohnesorge numbers ($Oh_o = Oh_d = 0.05, 0.07, 0.1$). The drop in scenario (a) ($Oh = 0.05$) undergoes a boundary stripping breakup mode. The drop with $Oh = 0.08$ in scenario (b) exhibits a similar evolution to the drop in scenario (a), although the deformation rate is slightly reduced and the front edge of the drop becomes flat. The central portion of the drop still retains a significant amount of fluid. In contrast, the central portion of the drop in scenario (c) is completely drained, resulting in the formation of a backward-facing bag, forming a very thin film of fluid near the symmetry axis.

The aspect ratios and centroid velocities are presented in Fig. 7 as functions of the dimensionless time t^* for the drops shown in Fig. 6. The decreasing trends of aspect ratios are similar since these three drops exhibit relatively comparable deformations compared to those across different Eötvös numbers (Eo). The initial slopes of



(a) Aspect ratio

(b) Centroid velocity

Figure 5: Effects of Eo number. The aspect ratios and the centroid velocities versus non-dimensional time shown in Fig. 4 with different Eo numbers: $Eo = 6, 12, 28.8$.

the velocity curves are identical across different Oh values. However, at later times, the centroid velocities decrease to varying degrees due to the differing extents of drop deformation. For $Oh = 0.05$ and $Oh = 0.07$, the drops experience boundary stripping breakup, resulting in a rapid decrease in centroid velocity. When Oh increases to 0.1, the drop forms a backward-facing bag with a very thin film of fluid near the symmetry axis, leading to an relatively slower decrease in centroid velocity.

3.4 Comparison of Results with Published Data

These results are compared with the data in Ref. [1], which investigates the effects of Eötvös (Eo) and Ohnesorge (Oh) numbers on falling drops within an axisymmetric computational domain. The contours of drops at varying Eo values are presented in Fig. 8, and those at different Oh numbers are shown in Fig. 9. Although the computational setups differ—my simulations involve two-dimensional (2D) falling drops with four non-slip boundaries, whereas Ref. [1] employs three-dimensional (3D) axisymmetric computational domains—the deformation trends with respect to Eo and Oh are consistent.

Specifically, for lower Eo values ($Eo = 6$ in my study and $Eo = 12$ in the referenced paper), the drop flattens at the rear while the front remains rounded. As Eo increases, the drop's rear becomes progressively more convex, gradually transforming into a backward-facing bag ($Eo = 12$ in my study and $Eo = 28.8$ and $Eo = 36$ in the paper). At higher Eo values ($Eo = 28.8$ in my study and $Eo \geq 60$ in the paper), the perimeter of the drop bends downstream, resulting that the central region of the drop retains a convex shape, and the thickness along the symmetry axis ceases to diminish.

Regarding the effects of the Ohnesorge number, given that $Eo = 28.8$ is a relatively large value in our study, we compare our results with those at $Eo = 144$ in the paper. When the Ohnesorge number is small ($Oh = 0.05$ in both studies), the drops undergo a boundary stripping breakup mode. As the Ohnesorge number increases ($Oh = 0.1$ in my study and $Oh = 0.125$ in the paper), the central portion of the drop is completely drained, resulting in the formation of a backward-facing bag and a very thin film of fluid near the symmetry axis.

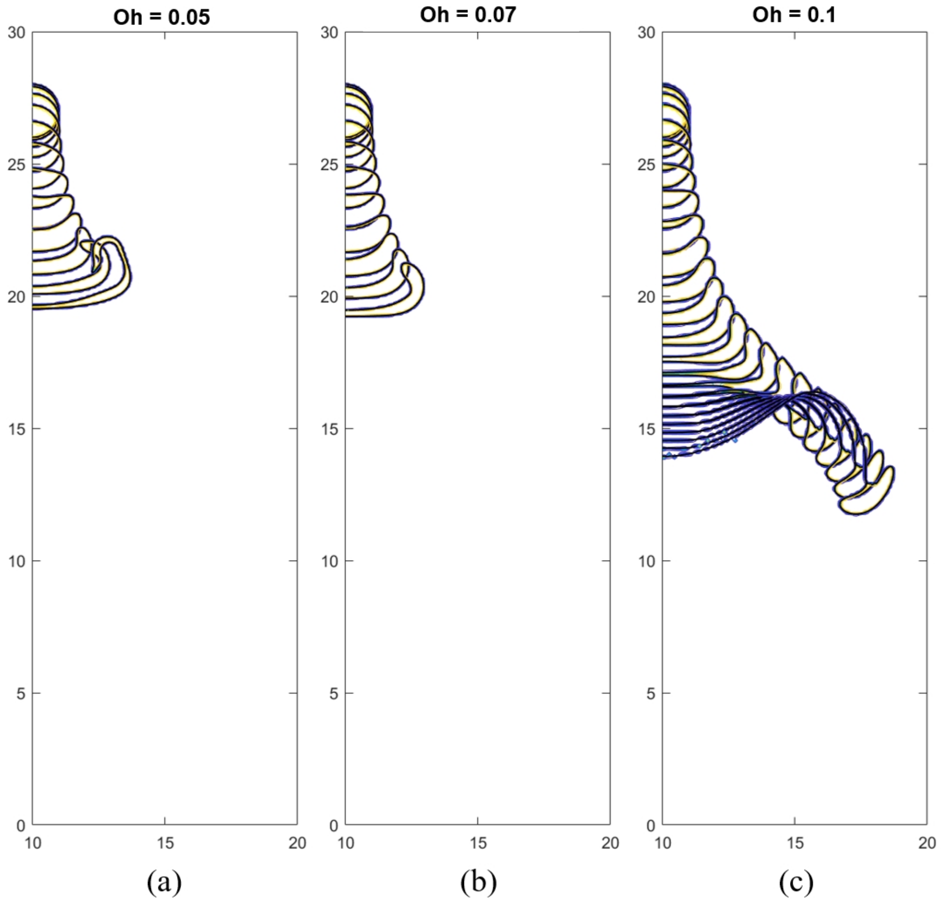


Figure 6: Effects of Oh on the falling and breaking up of a drop. Eo is fixed at 28.8. The time interval is $\Delta t^* = 0.3369$. The last interface is plotted at $t^* = 4$ (a), $t^* = 4$ (b) and $t^* = 8$ (c).

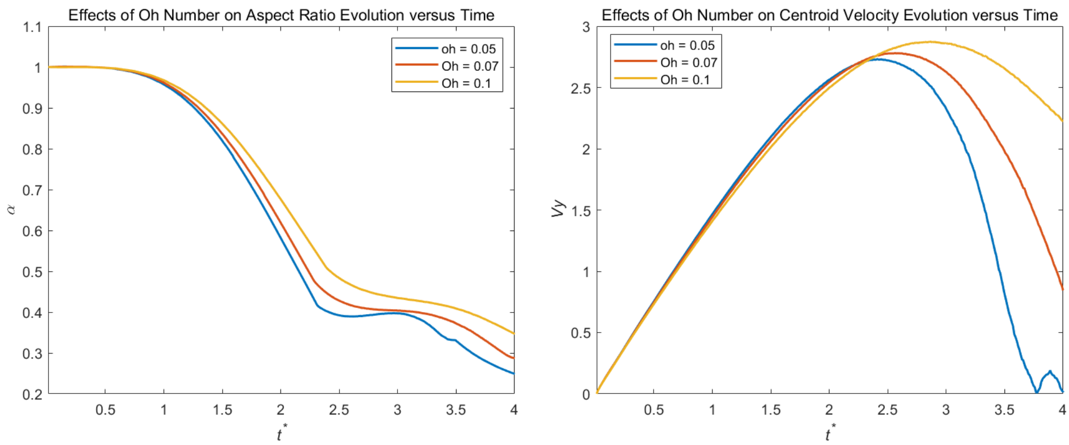
4 Conclusions

This study provides a detailed numerical investigation into the dynamics of a two-dimensional falling droplet under body forces. By employing the *CodeC3-frt-st-RK3.m*, we successfully simulated the droplet's behavior, focusing on the evolution of its aspect ratio and centroid velocity over time. The analysis highlighted the significant roles of the Eötvös (Eo) and Ohnesorge (Oh) numbers in governing droplet deformation and breakup modes. From this study, we obtain some interesting findings.

At low Eo values, droplets exhibit minimal deformation, maintaining a nearly spherical shape. As Eo increases, droplets undergo progressive deformation, transitioning through various breakup modes such as disk-like formation and backward-facing bag structures.

Lower Oh values facilitate rapid centroid velocity decreases due to boundary stripping breakup modes. Higher Oh values result in the formation of backward-facing bags with thin fluid films near the symmetry axis, affecting the droplet's velocity dynamics.

The numerical simulations effectively captured the complex interactions between body force, surface tension, and viscous forces governing the behavior of falling droplets.



(a) Aspect ratio

(b) Centroid velocity

Figure 7: Effects of Oh number. The aspect ratios and the centroid velocities versus non-dimensional time shown in Fig. 6 with different Oh numbers: $Oh = 0.05, 0.07, 0.1$.

The dependence of droplet deformation and breakup on Eo and Oh provides critical insights into predicting droplet stability and fragmentation under varying physical conditions.

Future research could extend this study by transitioning from 2D to 3D simulations to capture more intricate droplet dynamics and validate the findings in a more realistic spatial context. It is also valuable to explore a broader range of fluid properties, including varying density and viscosity ratios, to understand their combined effects on droplet behavior, investigate the influence of additional external forces, such as aerodynamic interactions or electromagnetic fields, on droplet dynamics and conduct experimental studies to further validate the numerical results and refine the simulation models.

Despite the comprehensive analysis, several open questions remain: What role does ambient turbulence play in the stability and fragmentation of falling droplets? How do variations in interfacial properties, such as surfactant presence, affect droplet behavior under similar conditions? How can the insights gained from these simulations be scaled to larger or smaller droplet sizes relevant to industrial applications?

Addressing these questions will further enhance the understanding of droplet dynamics, paving the way for advancements in fields ranging from meteorology to industrial processing.

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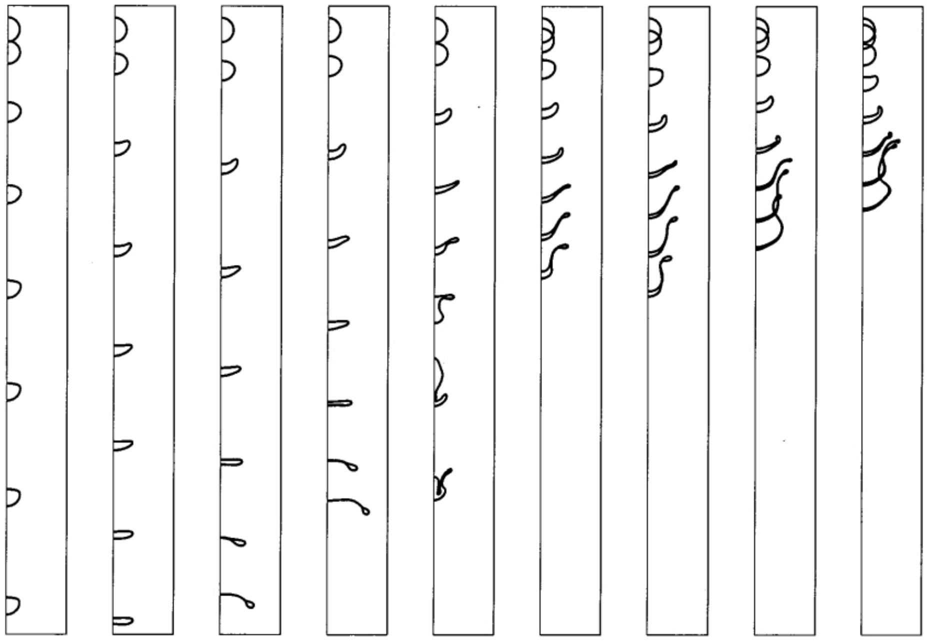

(a) $Eo = 12$ (b) $Eo = 24$ (c) $Eo = 28.8$ (d) $Eo = 36$ (e) $Eo = 48$ (f) $Eo = 60$ (g) $Eo = 72$ (h) $Eo = 96$ (i) $Eo = 144$

Figure 8: Effect of Eo on the deformation of drops at small Oh number in Ref. [1] with $\rho_d/\rho_o = 10$. $Oh_o = Oh_d = 0.05$. The gap between two successive drops in each column represents the distance the drop travels at a fixed time interval and the last interface is plotted at $t^* =$ (a) 11.19; (b) 15.82; (c) 14.85; (d) 13.83; (e) 11.19; (f) 7.15; (g) 7.83; (h) 6.78; (i) 5.54.

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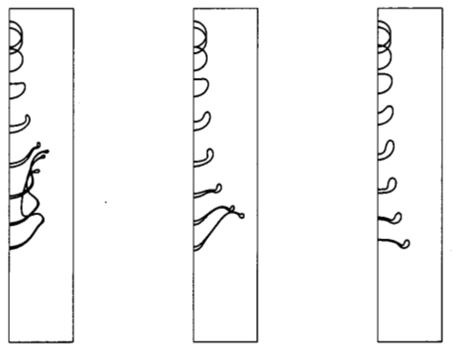

(d) $Oh = 0.05$ (e) $Oh = 0.125$ (f) $Oh = 0.25$

Figure 9: Effect of Oh for drops at large Eo number in Ref. [1] with $\rho_d/\rho_o = 10$. The drop evolution is shown for three $Oh_o = Oh_d = 0.05$, 0.125, and 0.25. Eo is 144 and Δt_p^* is 0.791.